

GenAI Developer Handbook (Chapter 1)

This chapter is written for interview preparation. It explains the core foundations required before learning RAG, AI Agents, Graph RAG and production GenAI systems.

AI vs ML vs DL vs GenAI

Artificial Intelligence (AI) is the broad goal of building intelligent systems. Machine Learning (ML) is a subset where systems learn patterns from data. Deep Learning (DL) uses deep neural networks to learn complex representations. Generative AI goes beyond prediction and classification—it generates new content such as text, code, images, audio and video.

Why Generative AI became popular

Large datasets, Transformer architecture, powerful GPUs and large-scale training enabled modern LLMs such as GPT-family models. These models learn language patterns and can perform many tasks from one base model.

What is an LLM?

A Large Language Model predicts the next token repeatedly. During training it sees enormous amounts of text and learns grammar, facts, reasoning patterns and programming syntax.

Tokens

LLMs read tokens instead of characters. A token may be a word, part of a word or punctuation. Context windows are measured in tokens.

Transformer Architecture

Pipeline: Input Text -> Tokenizer -> Embeddings -> Positional Encoding -> Multi-head Self Attention -> Feed Forward Network -> Output Token. Self-attention allows every token to understand the importance of every other token in the sentence.

Embeddings

Embeddings convert text into dense vectors. Similar meanings produce nearby vectors. They are the foundation of semantic search and Retrieval-Augmented Generation.

Hallucination

A hallucination is an incorrect answer generated with confidence. It happens because an LLM predicts likely tokens rather than verifying truth. RAG and grounding reduce hallucinations.

Interview Questions

1. Explain AI, ML, DL and GenAI. 2. Why are transformers better than RNNs? 3. What are tokens? 4. What is a context window? 5. What are embeddings? 6. Why do LLMs hallucinate? 7. Explain self-attention in simple words. 8. Why is positional encoding

required?

Quick Revision

Remember: • LLMs predict the next token. • Embeddings represent meaning as vectors. • Transformers use self-attention. • Context window limits how much information the model can process at once. • Hallucinations can be reduced using RAG.